Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/12/2010
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Project Title:	Radiation Effects on Bone Tissue and Cells in Reduced Gravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI formerly at Clemson University when NSBRI Postdoctoral Fellow Feb 2008-Oct 2010 (Ed., 12/18/2014)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-02 Postdoctoral Fellowships
Start Date:	02/01/2008	**End Date:**	09/30/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2010 (from 1/31/2011) per NSBRI (8/2010) NOTE: End date changed to 1/31/2011 (from 1/31/2010) per NSBRI (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bateman, Ted (MENTOR/Clemson University)		
Grant/Contract No.:	NCC 9-58-PF01403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

POSTDOCTORAL FELLOWSHIP
The spaceflight environment presents numerous challenges to astronauts' bone health, including the bone-debilitating effects of micro/reduced gravity. A reduction in bone volume and strength among astronauts represents a serious, mission-critical biomedical problem. Preliminary data and published accounts indicate that exposure to doses as low as 1 Gy of multiple types of ionizing radiation results in trabecular bone loss. However, it is unclear how bone already compromised by microgravity or partial gravity will be affected by exposure to space radiation.

Our intent is to enhance the understanding of the effects of radiation on bone health, particularly within a reduced gravity environment. Our ultimate goal is to develop appropriate countermeasures for bone loss in astronauts during exploratory missions. The proposed Aims will analyze the effects of modeled space radiation using scenarios applicable for Lunar Outpost missions.

Specific Aim 1: Examine the response of bone to a combination of spaceflight-associated challenges, including radiation exposure together with reduced limb bone loading.

Approach Aim 1: Skeletally mature female C57BL/6 mice, a strain sensitive to disuse-associated osteoporosis, were exposed to a 1 Gy dose or protons at Loma Linda Medical Center followed by a 4-week period of hindlimb suspension.

Main Findings Aim 1:

1) The combination of radiation and hindlimb suspension resulted in greater bone loss and deterioration of trabecular microarchitecture than the two challenges individually. Although both skeletal challenges induced atrophy, bone loss due to suspension alone was substantially greater than irradiation alone.

2) While reduced loading on the bone combined with microarchitecture resulted in excessive bone loss relative to either one challenge independently, the causal mechanisms for the stimuli may be independent. A thorough histological investigation was undertaken. Throughout the study, osteoclasts appear to be more active as an early response in both irradiated groups (irradiated only and irradiated + suspension) than non-irradiated groups.

Specific Aim 2: Examine the mechanistic causes of changes in bone at the cellular and tissue level following the individual and combined effects of radiation and unloading on mouse hindlimbs.

Approach Aim 2:

Year 1 of Funding: 13-week old C57BL/6 mice received whole body radiation with 2 Gy X-rays. X rays were chosen for these initial studies, as the relative biological effect of X-rays and protons are similar and bone loss from a 1 Gy dose has been shown to be similar (-13-15%) for both types at one month (preliminary data).

Year 2 of Funding: In an attempt to expand this NSBRI-funded developing technology to develop countermeasures for spaceflight and clinical applications, 16-week old mice were exposed to a 2 Gy dose of X rays. Some irradiated animals were given a commercially available osteoporosis medication (risedronate) to try to prevent any changes. Bone was examined at week 1, 2, and 3.

To apply this technology to clinical applications, the right hindlimbs of twenty-week old Sprague-Dawley rats received a dose of radiation modeling what the human hip receives during cancer radiotherapy. Animals were sacrificed at 2, 4, and 6 weeks to identify if modeled cancer therapy could induce bone loss. An antiresorptive agent (a commercially available bisphosphonate risedronate) was applied to try to prevent these changes.

Main Findings Aim 2:

1) Radiation appears to increase osteoclast number and activity as an early, acute response to radiation leading to early atrophy. Circulating markers of bone resorption are increased by 24 hours of exposure, with significantly increased osteoclast numbers present along trabeculae by 3 days. Trabecular bone loss is significant and substantial by Day 7 after exposure and occurs at multiple skeletal sites (proximal tibia; distal femur; fifth lumbar vertebrae). Bone quantity remains lower than control at Day 14 and 21.

2) Low dose irradiation of mice resulted in substantial deterioration of by one week after exposure. These changes occurred as a result of increased osteoclast activity and active bone resorption. An agent that blocks osteoclast activity (risedronate) completely prevented these changes. Modeling cancer radiotherapy resulted in severe deterioration of trabecular bone in rats at all time points investigated. Applying risedronate completely prevented deterioration of bone early after modeled cancer radiotherapy.

While Year 2 of funding was largely devoted to the development of a therapy for radiation-induced bone loss in both the spaceflight and clinical environment, identification of the causal mechanisms for bone loss must be identified. This past year we have clearly identified that radiation somehow turns on osteoclasts, and that this stimulation of active bone resorption (as opposed to suppressed bone formation) leads to bone loss. Our lab is identifying that markers of inflammation may be turning on these osteoclasts early after exposure. These include tumor necrosis factor-alpha (TNFa) and interleukin-1 (IL-1) activity. Therefore, for the coming year, we will investigate the role of these cytokines in radiation-induced bone loss. We will use a mutant mouse strain (reduced-inflammatory strain) developed in the lab of Dr. Albert Fornace, Jr. (Georgetown University) in collaboration with his NSBRI postdoctoral fellow, Dr. Daniella Trani. These mice will be exposed to a series of particle radiation exposures (both protons and iron nuclei; at low doses) at Brookhaven National Laboratory to identify if reducing inflammation can prevent bone loss.

Additionally, we will more appropriately model the skeletal effects of radiation during realistic spaceflight scenarios by irradiating mice with 0, 1, or 2 Gray dose of protons at a prolonged dose rate, modeling the maximum dose rate at the peak of a solar flare.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	The notion that radiation can increase osteoclast activity early after exposure leading to bone loss is a novel cause for radiation-induced osteoporosis. A radiation-induced increase in osteoclast function represents a potential target to prevent late-observed bone loss and fractures among cancer survivors. Our findings suggest that: 1) radiation increases osteoclast activity and number in during the first week after exposure, leading to substantial loss of bone; 2) the bone loss that occurs during the first week and any other evidence of deterioration can be prevented by applying a bisphosphonate as antiresorptive countermeasure, and 3) these observations are true in both mice (after low-dose photon exposures) and rats (after receiving targeted and modeled cancer radiation-therapy). Radiation-induced bone damage is a concern for radiation oncologists. Improvements in cancer treatment and diagnosis have led to increased long-term cancer survivorship. For example, of the more than 200,000 men who are diagnosed with prostate cancer each year, the 10 year survival rate approaches 90%. However, this growing population of long-term survivors is at risk of developing deleterious effects in normal tissues resulting from the use of therapeutic radiation. Skeletal complications, particularly fractures, are a recognized late-radiation induced effect. Fractures of the hip, ribs, clavicle, and humerus have documented in patients receiving targeted radiotherapy for pelvic and breast tumors. These bones absorb radiation during cancer treatment. The increased incidence of hip (particularly the femoral neck) fractures in postmenopausal women after receiving pelvic cancer treatment is substantial and can have severe negative impacts on a patient's mobility, independence, and survival. More post-menopausal women who fracture a hip will die within a year of the fracture than will die as a result of breast cancer. Therefore, preventing hip fractures due to radiation therapy (as well as vertebral breaks) are important in maintaining a patient's quality of life and health. Work from this project has identified that by preventing osteoclast activity after radiation exposure, bone loss can be prevented. This work has helped lead to the development of several clinical trials within oncology departments in California and North Carolina. If these observations translate into the clinic, it is possible that the notion of preventing bone deterioration resulting from cancer therapy by targeting the osteoclast can improve patient quality of care.
Task Progress:	Substantial progress was made during the course of the past year in terms of identifying bone loss appears to occur after exposure to ionizing radiation. From our animal models, we have clearly identified that radiation somehow stimulates bone loss by activating osteoclasts (cells responsible for bone resorption) within the first few days of exposure. This represents a new and novel cause for radiation-induced osteoporosis, and may contributed to the deterioration and fractures associated with cancer radiation therapy. Additionally, we have identified that by blocking the osteoclast with the use of a commercially available osteoporosis medication, we can prevent any changes in bone. Thus, we are developing a potential countermeasure for radiation-induced bone loss. Specific Aim 2: Examine the cause of radiation-induced bone loss and test preventative countermeasures (antiresorptive; anti-inflammatory) to prevent any early response that can lead to increased osteoclast activity. Progress: 16-week old C57BL/6 mice received whole body radiation with 2 Gy X-rays. This low dose of radiation caused a very quick loss of bone volume and deterioration of architecture in mice: virtually all indices of bone volume and architectural integrity were lower than control by one week after exposure. Bone loss and deterioration of microarchitecture was present in the proximal tibia, distal femur, and lumbar vertebrae. This osteoporosis occurred as a result of increased osteoclast number and activity, which was elevated only during the first week (which was when the majority of bone loss occurred). Additionally, bone formation rates were unchanged during the first week, so that loss of bone cannot be attributed to suppressed bone formation. However, by applying an agent that decreased osteoclast activity (the bisphosphonate risedronate, an osteoporosis medication) we completely prevented deterioration of bone after irradiation. These results were accepted for publication by the journal Bone in September of 2009, and are currently in press. Additionally, we attempted to determine if modeling the dose of radiation a human hip receives during radiation therapy for pelvic cancers (which increases the chance of hip fracture) in a rat model can induce bone loss Right hind limbs were exposed to 4 doses of 4 Gy at Wake Forest Comprehensive Cancer Center. Animals were sacrificed at 2, 4, and 6 weeks. Irradiation resulted in deterioration of the exposed bone relative to the non-irradiated contralateral limb. Reduced connectivity and a decreased number of more rod-like trabeculae were present . Deterioration was entirely prevented with risedronate administration. Thus, we have identified a potential target to prevent bone loss and fractures among cancer radiation therapy survivors.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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Patents	65,072. Patent, January 2010. Jan-2010 Bateman TA, Willey JS. "Use of osteoclast inhibiting compounds to prevent ionizing radiation-induced bone loss."